AI-BASED SMART SURVEILLANCE

ESP32-CAM + YOLO Object Detection + Flask Dashboard

WORKING MODEL
Real-time, explainable alerts

PROBLEM Conventional cameras record incidents but do not immediately identify restricted-zone entry, loitering, or crowding.

OBJECTIVE Build an affordable live surveillance system that detects people and objects, scores observable events, and supports human review.

PRINCIPLE & COMPONENTS

- ESP32-CAM provides a compact Wi-Fi MJPEG camera stream.
- Laptop/PC runs YOLO for object and person detection.
- Flask dashboard presents annotated video, score, status, and events.
- Optional: PIR sensor, buzzer, LED, microSD snapshots.

WORKING MECHANISM

- Capture: ESP32-CAM streams QVGA frames over local Wi-Fi.
- Analyze: YOLO finds people and common objects; a centroid tracker maintains identities.
- Evaluate: restricted-zone entry, dwell time, and crowd count update the score.
- Respond: dashboard labels NORMAL, MONITOR, or SUSPICIOUS and saves snapshots.

RESULTS & INNOVATION

- Live bounding boxes, class names, confidence, and people count.
- Orange restricted zone makes the rule visible and adjustable.
- Transparent scoring uses measurable events - not judgments about appearance.
- Low-cost architecture: ESP32-CAM for capture, PC for AI inference.

SYSTEM WORKFLOW

ESP32-CAM — Wi-Fi stream → **YOLO AI** — Objects + people → **RULE ENGINE** — Zone + loiter + crowd → **DASHBOARD** — Video + score → **ALERT / SNAPSHOT** — Human review

SUSPICION SCORING

+30	Restricted-zone entry
+30	Loitering beyond configured time
+20	Crowding at or above threshold

0-29 NORMAL

30-59 MONITOR

60+ SUSPICIOUS

APPLICATIONS

- Classrooms and laboratories
- Offices, shops, and warehouses
- Restricted rooms and entry points
- Prototype for smart-campus safety

CONCLUSION

A practical, affordable surveillance model that converts an ESP32-CAM feed into real-time, explainable decision support. Alerts are designed to prompt human review, not automated accusations.

Future scope: multi-camera tracking, secure alerts, and stronger event models.